AI 360: Deep Learning
Group Project Proposal

Project Title [EDIT]

Group: Group Name [EDIT]

Please describe your project in ≤4 pages excluding references. This template is for guidance, and you can decide the exact details to include, but we do require you to complete each of the tables and we strongly suggest keeping each section that we have provided in the template.

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1 Group Composition

Team Member	Student ID
Member 1	
Member 2	
Member 3	

2 Introduction and Problem Definition

Describe the problem that your project aims to address in 1 paragraph and explain why it is interesting or important. Your introduction should address the following questions:

- What problem are you trying to solve?
- What is the motivation and potential significance of this problem?
- What makes the problem suitable for a deep learning approach?
- What is the specific research question or hypothesis you intend to investigate?

Clearly define the scope of the project. State what you aim to demonstrate or achieve by the end.

3 Related Work and Background

Briefly discuss relevant prior work, existing methods, and approaches related to your problem. Identify the most relevant paper(s).

- How your proposed project relates to existing work.
- Which methods or ideas you intend to build upon.

4 Datasets

Describe the datasets you plan to use for training, validation, and testing. Specify:

- Dataset name and source (include a URL or citation).
- Task and type of data (e.g., images, video, text, or multimodal).
- Approximate dataset size and relevant characteristics.
- Intended training, validation, and test splits.
- Any preprocessing or data augmentation required.
- Access restrictions, licensing, or computational considerations.

Explain why the selected datasets are appropriate for the problem.

5 Available Hardware and Computational Resources

Describe the computational resources available to your group. Include relevant details such as GPUs or other accelerators available (model and memory, if known) beyond colab if any.

6 Proposed Model and Methodology

Describe your initial technical approach (this need not be a full definition of the model) and explain how you intend to develop and evaluate your deep learning solution.

Model and Starting Point. Specify the model or architecture (e.g., CNN, RNN, GNN, etc.) you intend to use and justify your choice. Clarify the starting point of your project:

- Will you use an existing pre-trained model?
- Will you fine-tune an existing model on your task?
- Will you train a model from scratch?
- Will you adapt or combine existing architectures?

Proposed Approach. Outline the main steps of your proposed method (As much as possible at this stage), including:

- Input representation and preprocessing.
- Model architecture and key components.
- Training objective or loss function, if applicable.
- Optimization strategy and important hyperparameters.
- Inference procedure and expected outputs.

We want you to provide enough detail to establish that the approach is technically plausible, while recognizing that the implementation and design may evolve.

Baselines and Evaluation.

Describe how you will determine whether your approach is successful. Specify:

- Baseline methods or reference models.
- Evaluation metrics appropriate to the task.
- Experimental comparisons or ablation studies.
- Expected results and how they will be interpreted.

7 Timeline and Deliverables

Provide a realistic timeline for the project, identifying milestones, deliverables, and dependencies.

Time Period	**Milestone / Activities**	**Expected Deliverable**
Week 1	Lit. review and problem refinement	Finalized problem and plan
Week 2	Dataset preprocessing	Working data pipeline
Week 3	Baseline implementation	Initial baseline results
Week 4	Model development and training	Initial model results
Week 5	Final results and report	Final report and slides

Adapt the timeline to the actual course schedule. Identify at least one intermediate milestone that can be used to assess feasibility and revise the approach if necessary.

8 Division of Work

Describe the planned responsibilities of each team member. Explain how team members will coordinate and integrate their work. Ensure that the distribution of work is clear and reasonably balanced.

Team Member	**Primary Responsibilities**	**Expected Contributions**
Member 1, 2, 3	Literature review and problem formulation	Related work and project design
Member 1	Dataset preparation and preprocessing	Data pipeline and analysis
Member 2	Model implementation and training	Model and training code
Member 3	Evaluation and experimental analysis	Metrics, comparisons, and visualization

References